

minimum-to-median wage (Kaitz) index. At the same time, the number of excess jobs also rose for these events to a nearly identical extent. As a consequence, there is no relationship between the employment estimate and the Kaitz index up to around 59%, confirming that the minimum wage changes in the United States that we study have yet to reach a level above which significant disemployment effects emerge.

The lack of responses in overall employment might mask some heterogeneity in response across types of workers. Our localized approach around the minimum wage can be easily applied to various subgroups, including those where only a small fraction of workers are affected by the minimum wage. As a result, we can provide a more complete picture of how various groups are affected by the minimum wage.

We examine whether there is a shift from low-skill to high-skill workers at the bottom of the wage distribution by partitioning workers into groups based on education and age. Comparing the number of excess jobs at or above the new minimum wage and missing jobs below it across age-by-education groups shows no evidence that low-skilled workers are replaced with high-skilled workers following a minimum wage increase. We also use demographics to predict the probability of being exposed to the minimum wage increase, and then assign workers to high, medium, and low probability groups along the lines of [Card and Krueger \(1995\)](#). While there is considerable variation in the bite of the policy, the employment effects in these subgroups are mostly close to zero and not statistically significant. The similar responses across demographic groups also suggest that the benefit of minimum wage policies were shared broadly.

Our approach also allows us to provide a more comprehensive picture on responses across various sectors of the economy. We show that the minimum wage is likely to have a negative effect on employment in the tradeable sector, and in manufacturing in particular—with an employment elasticity with respect to own wage of around -1.4 —although the estimates are imprecise. At the same time, the effect of the minimum wage is close to 0 in nontradeable sectors (such as restaurants or retail), which employ most minimum wage workers in the United States today. This evidence suggests that the industry composition of the local economy is likely to play an important role in determining the disemployment effect of the minimum wage ([Harasztsi and Lindner forthcoming](#)).

This article makes several contributions to the existing literature on minimum wages. First, it relates to the large and controversial literature on the employment effects of the minimum wage. The debate has often been concentrated on the impact on teen employment (Card 1992; Neumark and Wascher 1992; Neumark, Salas, and Waschler 2014; Allegretto et al. 2017), workers in specific sectors (Lester 1964; Katz and Krueger 1992; Card and Krueger 1994; Dube, Lester, and Reich 2010), or workers earning low wages prior to the minimum wage increase (Currie and Fallick 1996; Abowd et al. 2000; Clemens and Wither 2019), while the evidence on the impact on overall employment is scant. By disaggregating the standard difference-in-differences estimates by wage bins, we can identify the effects of the minimum wage on overall employment and obtain meaningful first-stage wage effects at the same time.

A notable exception studying overall employment changes is Meer and West (2016), who examine the relationship between aggregate employment at the state level and minimum wage changes without assessing the wage effects. Meer and West (2016) find a large negative employment estimate using variants of the classic two-way fixed effects regression on log minimum wage. To highlight the importance of disaggregating the aggregate employment effects into wage bins, we calculate the bin-by-bin employment effects in such a specification. This exercise produces a striking finding: the specifications that indicate a large negative effect on aggregate employment tend to be driven by an unrealistically large drop in the number of jobs at the upper tail of the wage distribution, which is unlikely to be a causal effect of the minimum wage. We also provide an explanation for why the classic two-way fixed effect and our event study approach produce different results. We show that the large negative effect on employment is driven entirely by inclusion of the 1980s and the early 1990s in the sample—a period with very few minimum wage changes. However, aggregate employment changes in the 1980s turn out to be correlated with minimum wage changes in the 2000s. Although inclusion of the 1980s biases the estimation in the two-way fixed effect approach, it does not affect our event study approach, which focuses on employment changes locally around the event window. It is worth noting that the disagreement on the choice of specification for estimating the impact of minimum wages on teen employment is also driven by these early period confounding shocks (Neumark, Salas, and Wascher 2014; Allegretto et al. 2017). We

find that in the post-1992 period, there is little evidence of disemployment for teens across any of the standard specifications.

Our article also contributes to the literature on the effect of the minimum wage on overall wage inequality (DiNardo, Fortin, and Lemieux 1996; Lee 1999; Autor, Manning, and Smith 2016). These papers examine shifts in the wage density and assume away any possible disemployment effect. In contrast, we focus on the frequency distribution of wages instead of the wage density, which allows us to assess the effect on wage inequality and employment at the same time.¹ We show that the measured wage spillovers are not an artifact of disemployment, which would truncate the wage distribution. In addition, we provide a wide range of evidence that these spillovers are unlikely to be an artifact of measurement error. Our spillover estimates are similar to the findings of Autor, Manning, and Smith 2016 and Brochu et al. (2017), and more limited than those in Lee (1999).

Finally, our article is also related to the literature that uses bunching to elicit behavioral responses to public policies (Kleven 2016). At the same time, while most bunching analyses estimate the counterfactual distribution from purely cross-sectional variation (Saez 2010; Chetty et al. 2013), here we use a difference-in-differences strategy to construct the counterfactual frequency distribution of wages and the estimated excess and missing jobs.

The rest of the article is structured as follows. Section II explains the conceptual approach and the empirical implementation. Section III presents the main empirical findings on overall employment effects, wage spillovers, and heterogeneous responses to the minimum wage. Section IV demonstrates the importance of assessing employment changes far above the minimum wage and highlights problems with the classic two-way fixed effects estimation. Section V concludes. Finally, all the Appendix materials can be found in the Online Appendix.

II. METHODOLOGY AND DATA

II.A. *The Conceptual Framework*

We infer the effect of the minimum wage from the employment changes at the bottom of the wage distribution. We illustrate our

1. In a recent working paper, Brochu et al. (2017) use the hazard rate for wages to estimate spillover effects in the presence of disemployment effects.

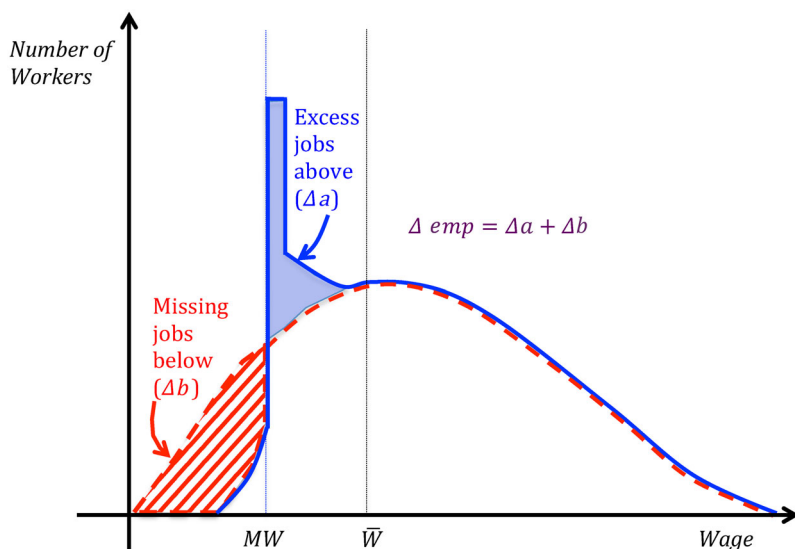


FIGURE I

The Impact of Minimum Wages on the Frequency Distribution of Wages

The figure shows the effect of the minimum wage on the frequency distribution of hourly wages. The red dashed line (color version available online) shows the wage distribution before the introduction of the minimum wage, and the blue solid line shows the distribution afterwards. Because compliance is less than perfect, some workers are paid below the minimum wage, and the post-treatment distribution starts below the minimum wage. For other workers, the introduction of the minimum wage produces “missing jobs” (Δb), as shown by the striped red shaded area (under the red dashed line) between the origin and MW . These missing jobs may either reflect workers getting a raise, or their jobs being destroyed. The former group creates the “excess jobs above” (Δa), as shown by the solid blue shaded area (under the blue solid line) between MW and $\bar{W}$, the upper limit for any effect of the minimum wage on the earnings distribution. The overall change in employment due to the minimum wage (Δe) is the sum of the two areas ($\Delta a + \Delta b$).

approach using Figure I, which summarizes the effect of the minimum wage on the wage distribution. The red dashed line (color version available online) shows a hypothetical (frequency) distribution of wages in the absence of the minimum wage. The blue solid line depicts the actual wage distribution with a minimum wage at MW .

In the presence of a binding minimum wage, there should be no jobs below MW . In practice, however, some jobs observed in the data will be sub-minimum wage because of imperfect coverage, imperfect compliance, or measurement error. Therefore, the